Computer Applications in Business

Lecture 5
Transforming Data into Information

Farjana Rahman Farin
Lecturer, Dept. of CSE, Sonargaon University

How Computers represent Data

Data representation in computer science is about how a computer interprets and functions with different types of information. Different information types require different representation techniques.

When working with various forms of data, it is important to grasp a fundamental understanding of:

- Binary system
- Bits and Bytes
- Number systems: decimal, hexadecimal
- Character encoding: ASCII, Unicode

- Data in a computer system is represented in binary format, as a sequence of 0s and 1s, denoting 'off' and 'on' states respectively. The smallest component of this binary representation is known as a bit, which stands for 'binary digit‘

- A byte, on the other hand, generally encompasses 8 bits. An essential aspect of expressing numbers and text in a computer system, are the decimal and hexadecimal number systems, and character encodings like ASCII and Unicode.
- For instance,

1101=13

This interpretation happens consistently in the background during all of your interactions with a computer system.

- 1 bit = Binary Digit
- 8 bits = 1 byte

- EBCDIC. EBCDIC (pronounced EB-si-dic) stands for Extended Binary Coded Decimal Interchange Code. EBCDIC b an eight-bit code that defines 256 symbols. It is still used in IBM mainframe and midrange systems, but it is rarely encountered in personal computers.
- ASCII. ASCII (pronounced AS-key) stands for the American Standard Code for Information Interchange. Today, the ASCII character set is by far the most commonly used in computers of all types.
- Unicode. The Unicode Worldwide Character Standard provides up to four bytes—32 bits—to represent each letter; number, or symbol. With four bytes, enough Unicode codes can be created to represent more than 4 billion different characters or symbols

How Computers Process Data

Input:

Data Entry Devices (e.g., Keyboard, Mouse): Users input data into the computer through various devices.

Processing:

Central Processing Unit (CPU): The CPU processes data by executing instructions stored in memory.

Control Unit: Manages the flow of data and instructions within the CPU.

Arithmetic Logic Unit (ALU): Performs arithmetic and logical operations.

Output:

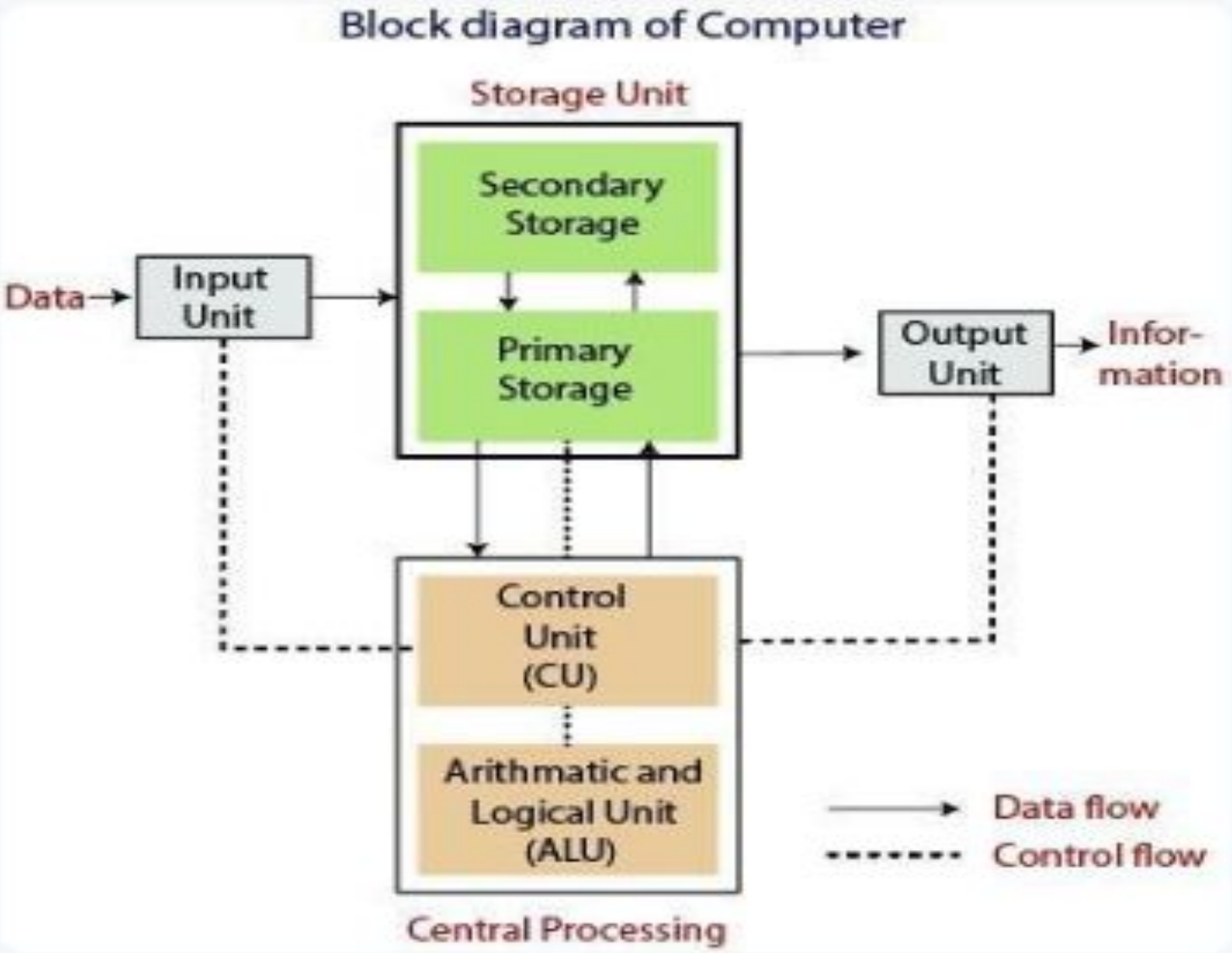
Output Devices (e.g., Monitor, Printer): Processed data is presented to users in a human-readable form.

Storage:

- **Memory (RAM):** Temporarily stores data and instructions for the CPU.
- **Storage Devices (e.g., Hard Drive, SSD):** Stores data persistently for long-term use.

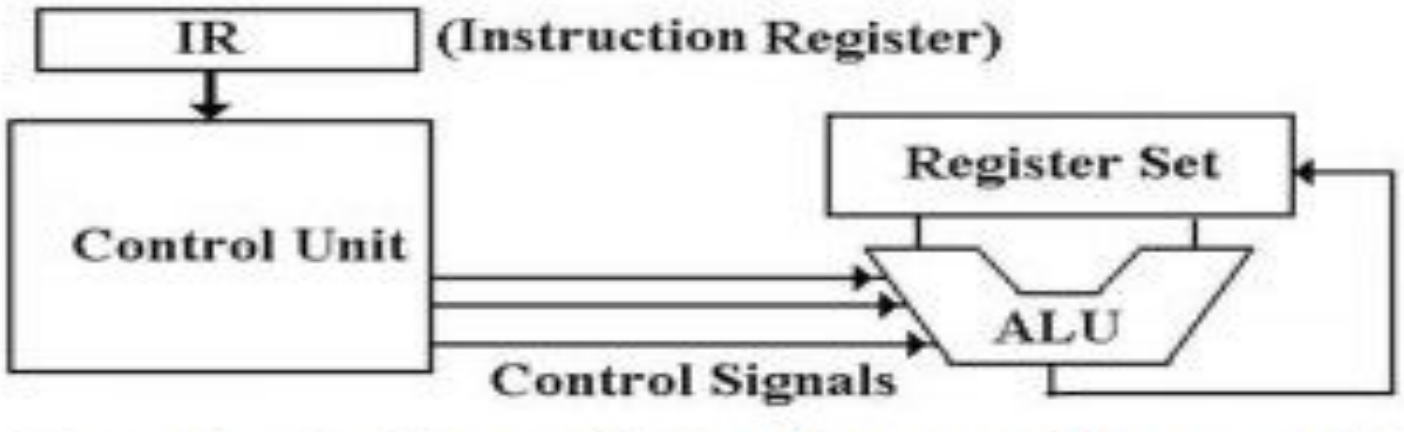
The flow of data starts with input, goes through processing by the CPU, results in output presented through output devices, and involves storage in both temporary (RAM) and long-term (storage devices) memory.

If you're looking for a visual representation, you might want to search for "information processing cycle" or "computer processing diagram" online to find illustrations that depict these concepts visually.


Block diagram of Computer
Storage Unit
Secondary Storage
Primary Storage
Data
Input Unit
Output Unit
Infor-mation
Control Unit (CU)
Arithmatic and Logical Unit (ALU)
Data flow
Control flow
Central Processing

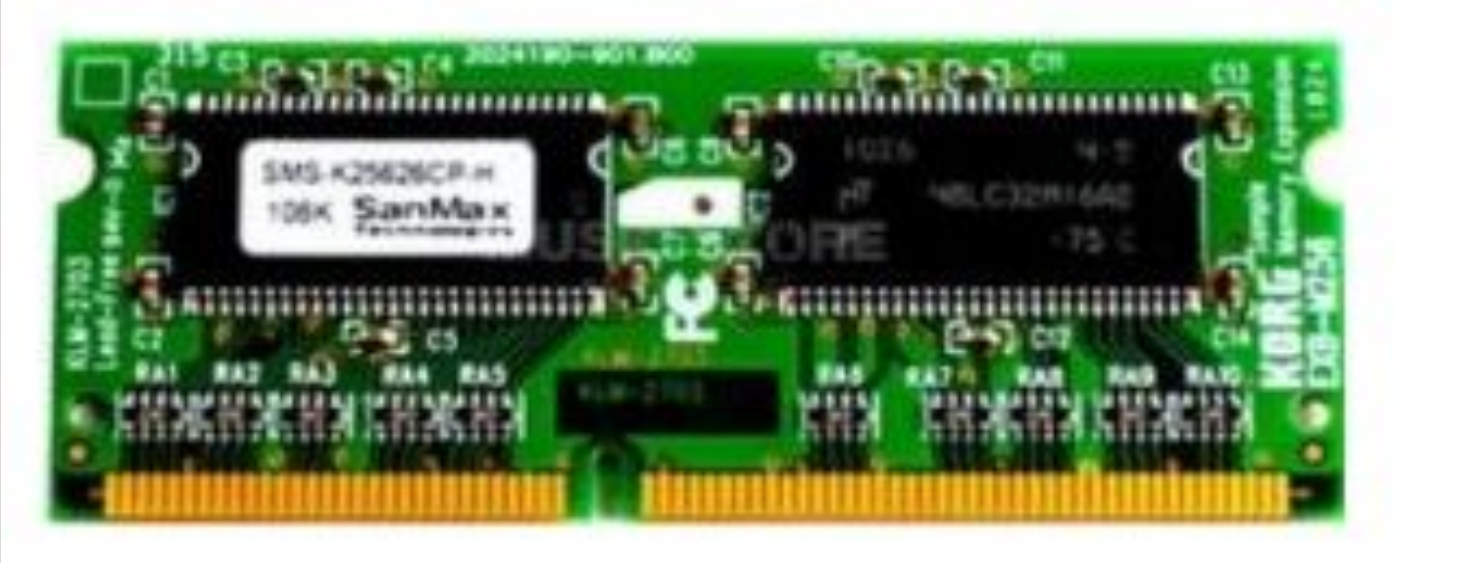
Factors Affecting Processing Speed: CPU's design determines its basic speed, but other factors can make chips already designed for speed work even faster.

1. **Registers:** The first factor that affects the processing speed of a computer is a register. A register is a small, high-speed memory inside a **CPU**. CPU contains a number of registers. The data and instructions are stored by registers while the CPU processes them.

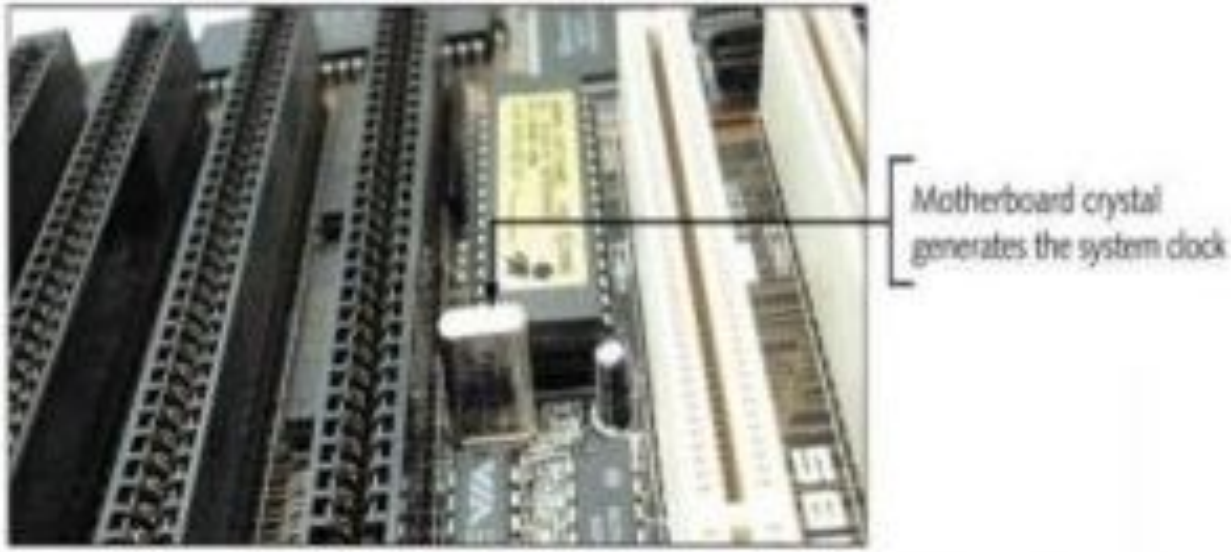


Computer Registers effect on the processing speed of a computer system

RAM: The size of RAM is also included in the factors that affect the processing speed of the computer. The bigger amount of RAM means that more program instructions and data can be stored in memory.

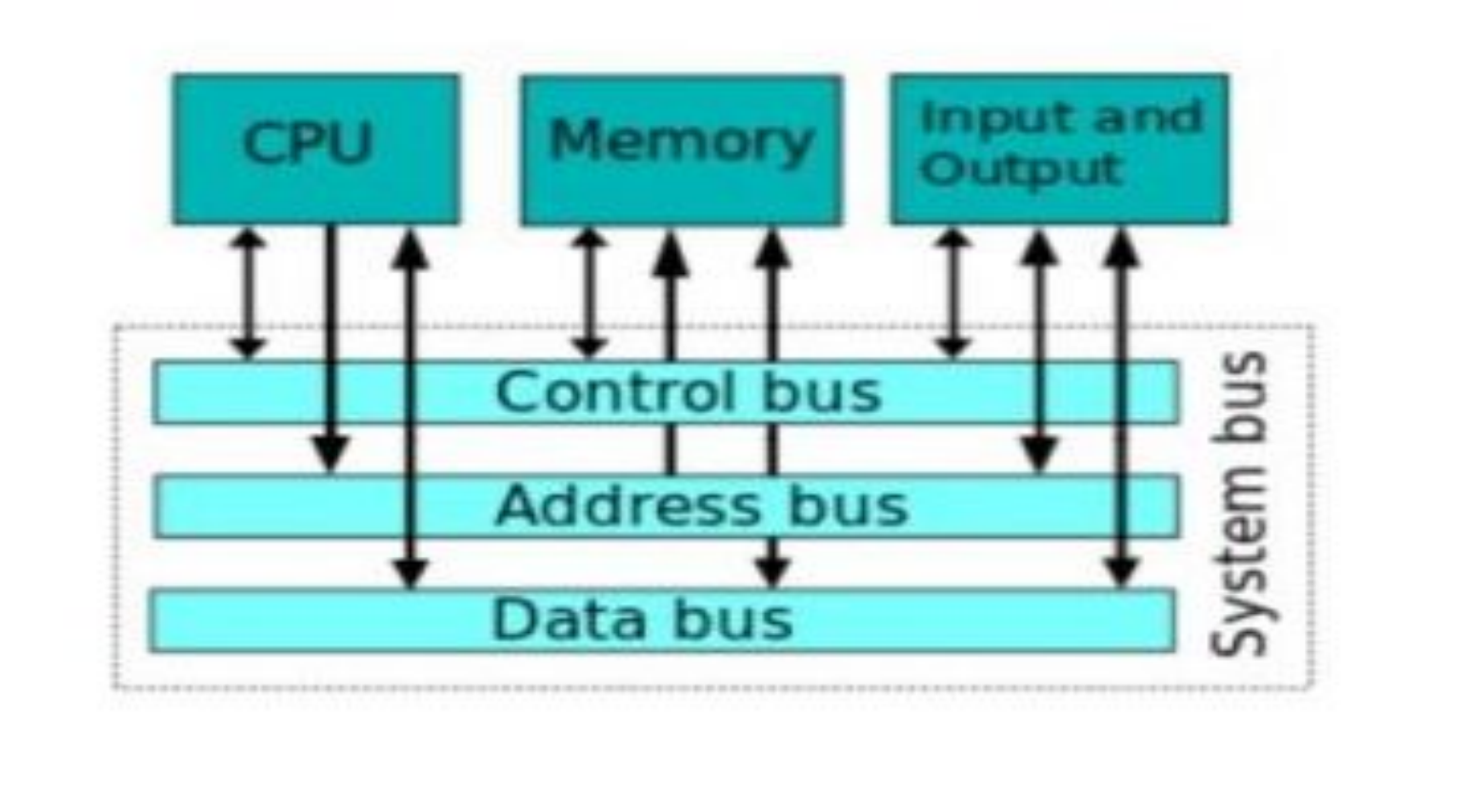


System Clock: A system clock is an electronic component that is an important factor that affects the processing speed of the computer. It generates electric signals at a fast speed. The time required to execute a single instruction is called clock speed. Clock speed tells us how much time a CPU takes to execute each instruction. CPU takes from 1 to 6 clock ticks to execute one instruction.

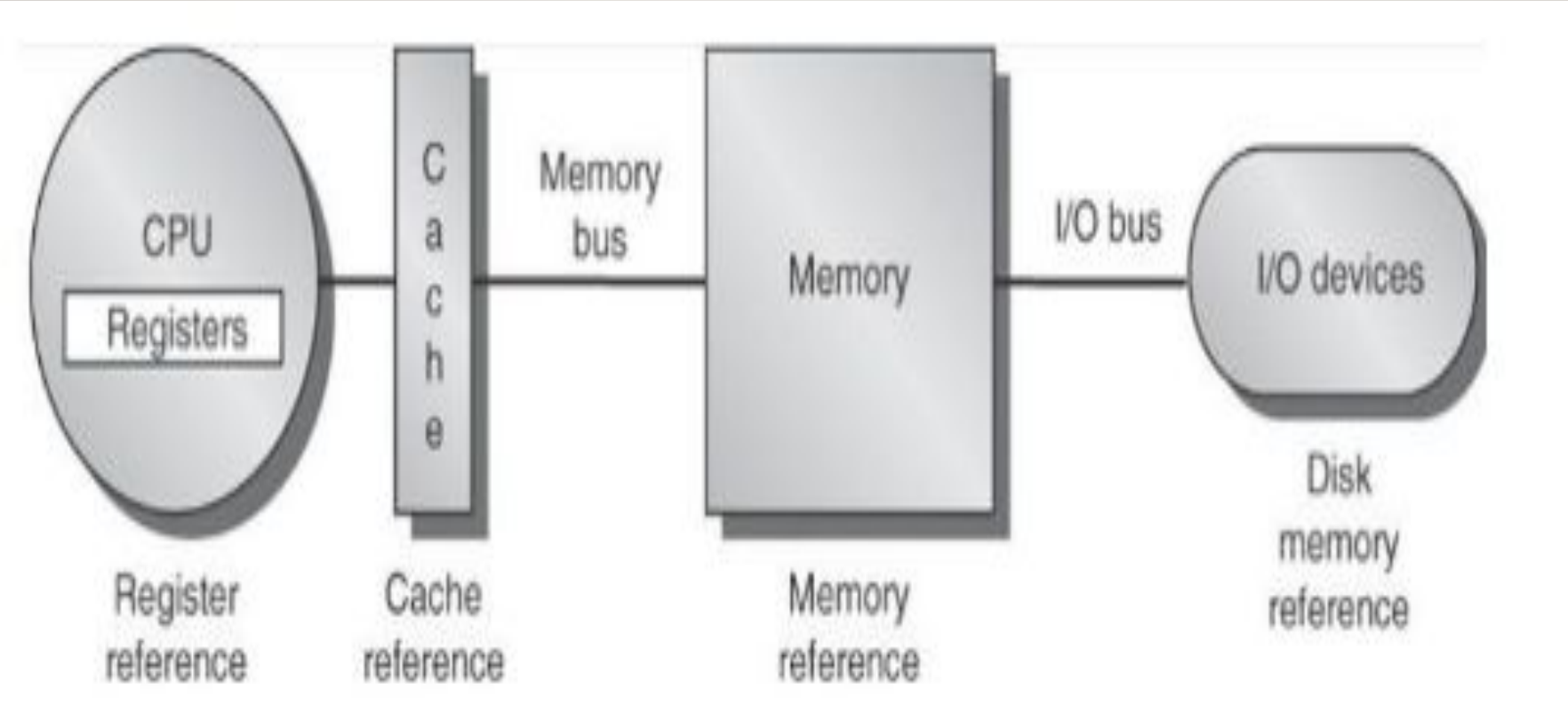


BUS:

The physical path between the components of a computer is a bus. Bus width is also considered in the factors that affect the processing speed of a computer. A higher bus width means that the user can carry more data. It increases the performance of the computer.



Cache Memory: A cache is a high-speed memory that holds the most recent data and instructions that have been loaded by the CPU. It is designed to speed up the transfer of data and instructions. The cache is located directly on the CPU and RAM. It is faster than RAM.



Core Count: The number of cores home is the number of the structures that our CPU can execute at the same time. For example, a quad-core CPU can execute four instructions at the same time because it's got four cores. Whereas a hex-core CPU can execute six instructions at the same time because it's got six cores.

Computer Applications in Business

Lecture 3 & 4
Interacting with Personal Computer & Ergonomics

Farjana Rahman Farin
Lecturer, Dept. of CSE, Sonargaon University

Devices used to interact with a Computer:

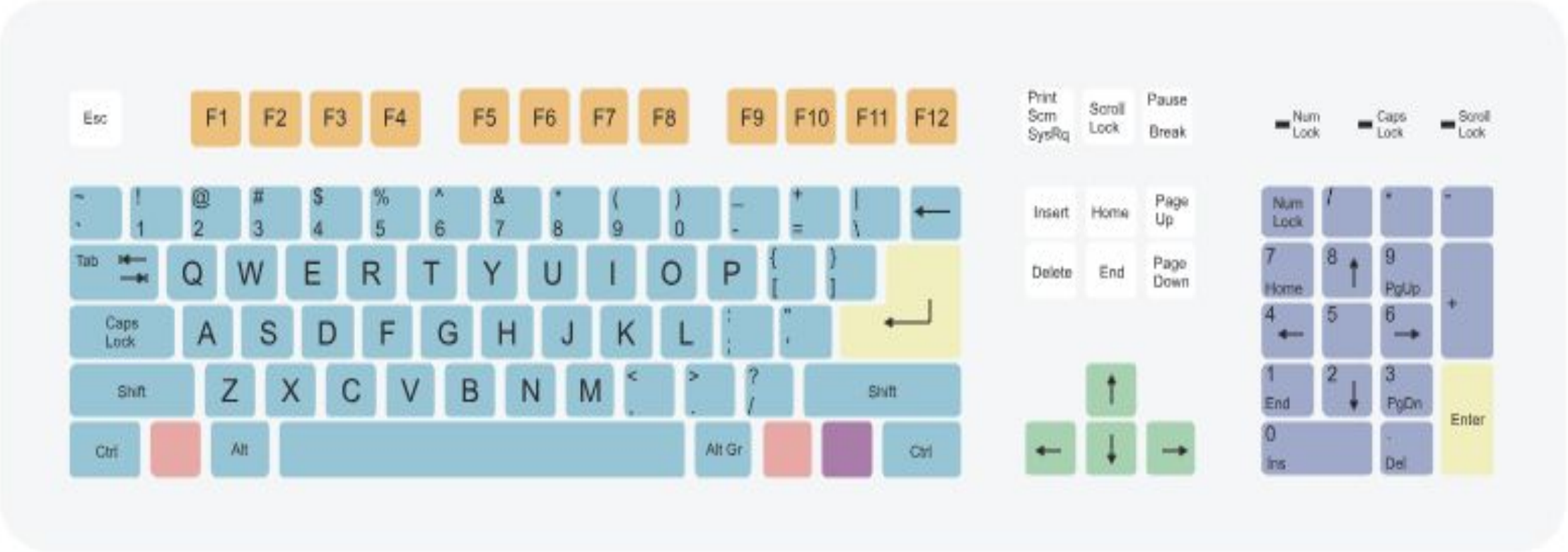
- Keyboard
- Mouse
- Touchpad

If you think of the CPU as a computer's brain, then you might think of the input devices as its sensory organs—the eyes, ears, and fingers. From the user's point of view, input devices are just as important as the CPU, perhaps even more important.

The Standard Keyboard Layout

A standard keyboard includes about 100 keys; each key sends a different signal to the CPU.
The keyboard was one of the first peripherals to be used with computers, and it is still the primary input device for entering text and numbers.

The standard keyboard has from 101 to 103 keys. The letters are in the Querty layout. Numbers in a row above, spacebar front and center.



Typewriter keys

Function keys

Enter keys

System keys

Numeric keypad

Other

Application key

Cursor control keys

The Alphanumeric Keys:

- The alphanumeric keys the area of the keyboard that looks like a typewriter's keys are arranged the same way on almost every keyboard. Sometimes this common arrangement is called the QWERTY (pronounced KWER-tee) layout because the first six keys on the top row of letters are Q, W, E, R, T, and Y.

The Modifier Keys:

- The SHIFT, ALT (Alternate), and CTRL (Control) keys are called modifier keys because they modify the input of other keys.

The Numeric Keypad :

The numeric keypad looks like a calculator's keypad, with its 10 digits and mathematical operators (+, -, *, and /). The numeric keypad also features a NUM LOCK key.

The Function Keys:

The function keys, which are labeled F I, fl, and so on. menus or dialog boxes. Each function key's purpose depends on the program you are using.

The Cursor-Movement Keys :

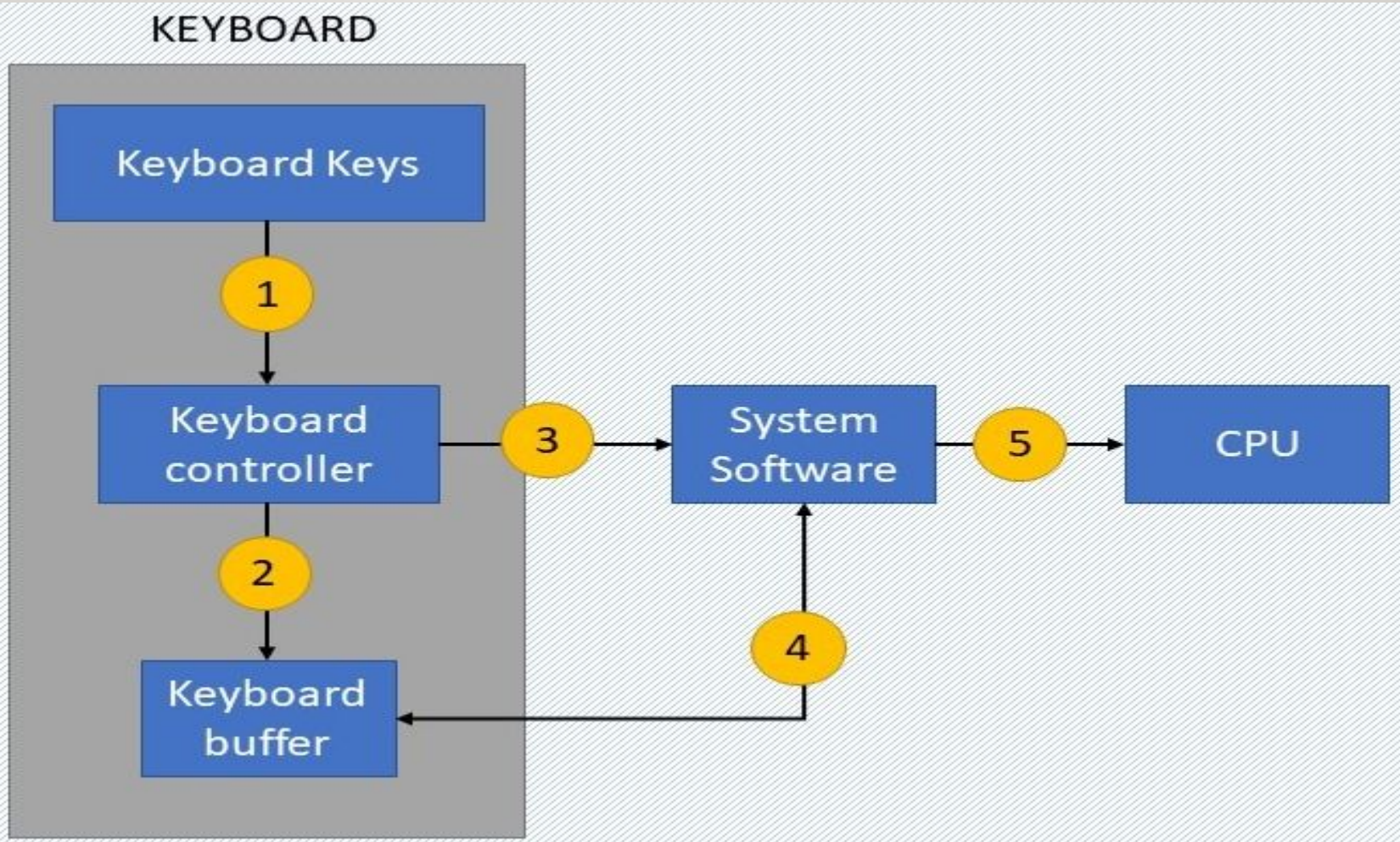
Most standard keyboards also include a set of cursor movement keys, which let you move around the screen without using a mouse. In many programs and operating systems, a mark on the screen indicates where the characters you type will be entered.

Special-Purpose Keys :

In addition to the five groups of keys described earlier, all IBM compatible keyboards feature six special-purpose keys, each of which performs a unique function.

- **Start**
- **shortcut**

How the Computer Accepts Input from the Keyboard



Using the Mouse

You use a mouse to move the pointer to a location on the screen, a process called pointing. Everything you do with a mouse is accomplished by combining pointing with these techniques:

» Clicking
» Double-clicking
» Dragging
» Right-clicking

Ergonomics and Input Devices

- **Ergonomics,** which is the study of the physical relationship between people and their tools—such as computers—addresses these issues.
- The term ergonomically correct means that a tool or a workplace is designed to work properly with the human body, and thus reduces the risk of strain and injuries.

Repetitive Stress Injuries: These ailments are called repetitive stress injuries (RSIs) or repetitive strain injuries and result from continuously using the body in ways it was not designed to work.

Here are some other tips that can help you avoid RSIs while working with your keyboard and mouse:

1. **Use an Ergonomic Keyboard :** An ergonomic keyboard allows you to hold your hands in a more natural position (with wrists straight, rather than angled outward) while typing.

2. Use a Padded Wrist Support :

If you type a lot, a wrist support can be helpful by allowing you to rest your hands comfortably when you are not actually typing.

3. Keep Your Wrists Straight :

When typing, your hands should be in a straight line with your forearms, when viewed either from above or from the side.

4. Sit Up Straight : Avoid crossing your legs in front of you or under your chair for long periods.

.

5.Learn to type : You will use the keyboard more efficiently and naturally if you know how to type.

6. Take Frequent Breaks : Get up and move around for a few minutes each hour, and stretch occasionally throughout the day.

.

Ergonomics and Monitor

Just as too much keyboarding or improper typing technique can lead to hand or wrist injuries, too much time at a monitor can endanger your eyesight. Protecting your eyesight means choosing the right kind of monitor and using it correctly.

Monitor tips:

- Adjust the monitor height so that the top of the screen is at—or slightly below—eye level.
- Your eyes should look slightly downward when viewing the middle of the screen.
- Position the monitor at least 20 inches (51 cm) from your eyes—about an arm's length distance. If your screen is larger, add more viewing distance.

- Keep glare down by adjusting the screen position.
- Maintain the distance between your screen-scanning eyes and your screen by tilting the monitor back 10° to 20°.
- If you wear bifocals (we'll never tell!), lower the monitor below eye level and turn your screen upward, tilting it back 30° to 45°.

Most input devices arc designed to be used by hand. Even specialized devices like touch screens enable the user to interact with the system by using his or her fingertips. Unlike keyboards and mice, many of these input devices are highly intuitive and easy to use without special skills or training.

- **Pen-based systems:** You hold the pen in your hand and write on a special pad or directly on the screen. You also can use the pen as a pointing device, like a mouse, to select commands by tapping the screen.

- Pen-based computers are commonly used for data collection, where the touch of a pen might place a check in a box to indicate a par that must be ordered or a service that has been requested

Touch Screens: Touch screens accept input by allowing the user to place a fingertip directly on the computer screen, usually to make a selection from a menu of choices. Most touchscreen computers use sensors on the screen's surface to detect the touch of a finger, bur other touch screen technologies are in use, as well.

- Touch screens have become common in fast-food restaurants, department stores, drugstores, and supermarkets, where they are used for all kinds of purposes, from creating personalized greeting cards to selling lottery tickets.

Game Controllers: You may not think of a game controller as an input device, but it is. Personal computers are widely used as gaming platforms, challenging dedicated video game units like the Sony PlayStation

Optical Input Devices:

- Bar Code Reader: Bar code readers are one of the most widely used input devices. The most common type of bar code reader is the flatbed model, which is commonly found in supermarkets and department stores.

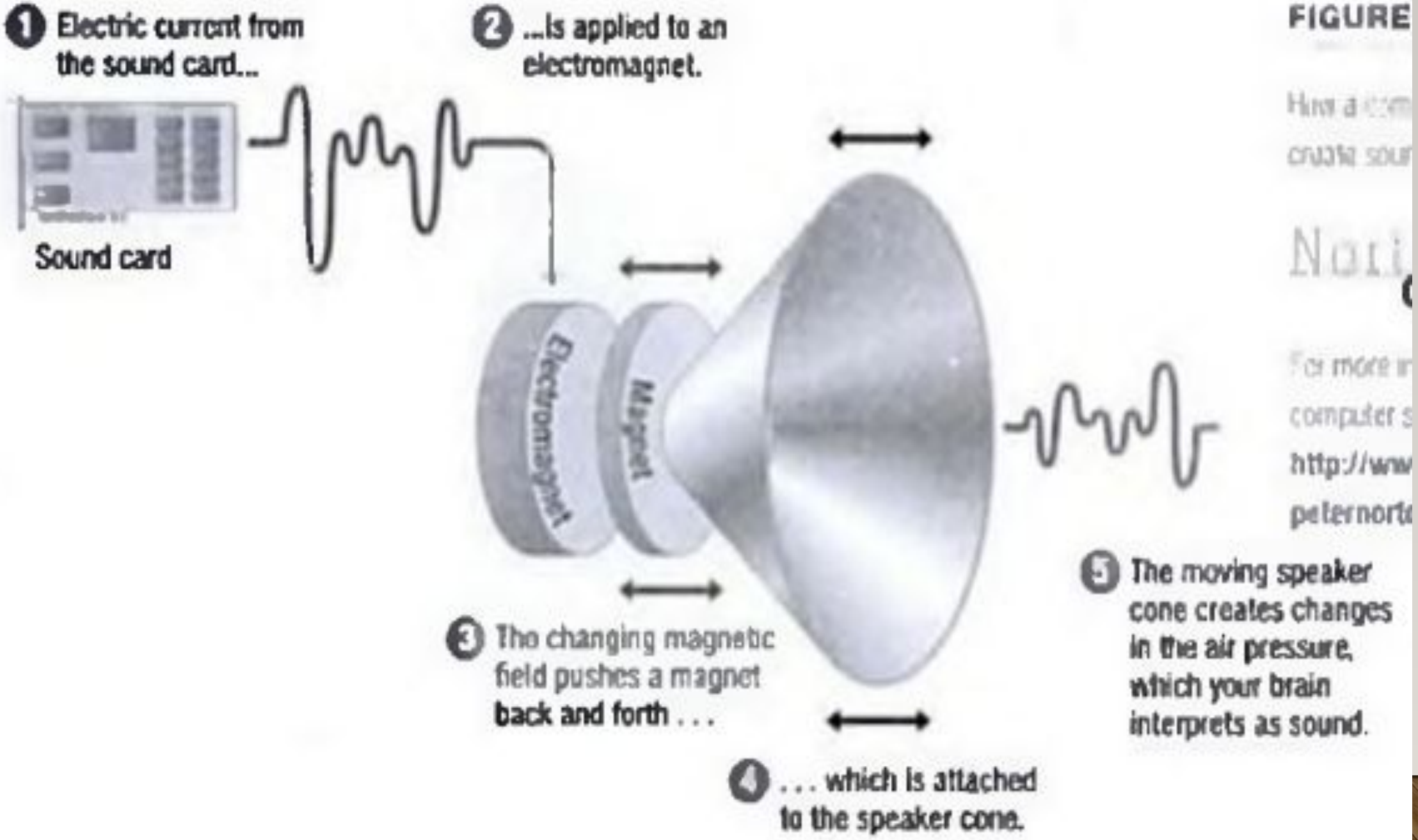
Workers for delivery services, such as FedEx, also use handheld bar code readers in the Held to identify packages

Video

- The quality of the images that a monitor can display is defined as much by the video card (also called the video controller or the video adapter) as by the Monitor itself.
- the video controller is an intermediary device between the CPU and the monitor. It contains the video-dedicated memory and other circuitry necessary to send information to the monitor for display on the screen.
- In most computers, the video card is a separate device that is plugged into the motherboard.
- The video controller contains its own on-board processor and memory, called video RAM

Sound

- The most complicated part of a computer’s sound system is the sound card.
- A computer's sound card is a circuit board that converts sound from analog to digital form, and vice versa, for recording or playback. A sound card actually has both input and output functions.
- your voice, for instance, you connect the microphone to the sound card’s input jack. Other audio input devices connect to the sound card as well, such as the computer's CD-ROM or DVD drive.
- The sound card translates digital signals into analog signals that drive the speakers.
- The sound card translates digital signals into analog signals that drive the speakers.


1 Electric current from the sound card...
2 ...is applied to an electromagnet.
Sound card
Electromagnet
Magnet
3 The changing magnetic field pushes a magnet back and forth . . .
4 . . . which is attached to the speaker cone.
5 The moving speaker cone creates changes in the air pressure, which your brain interprets as sound.

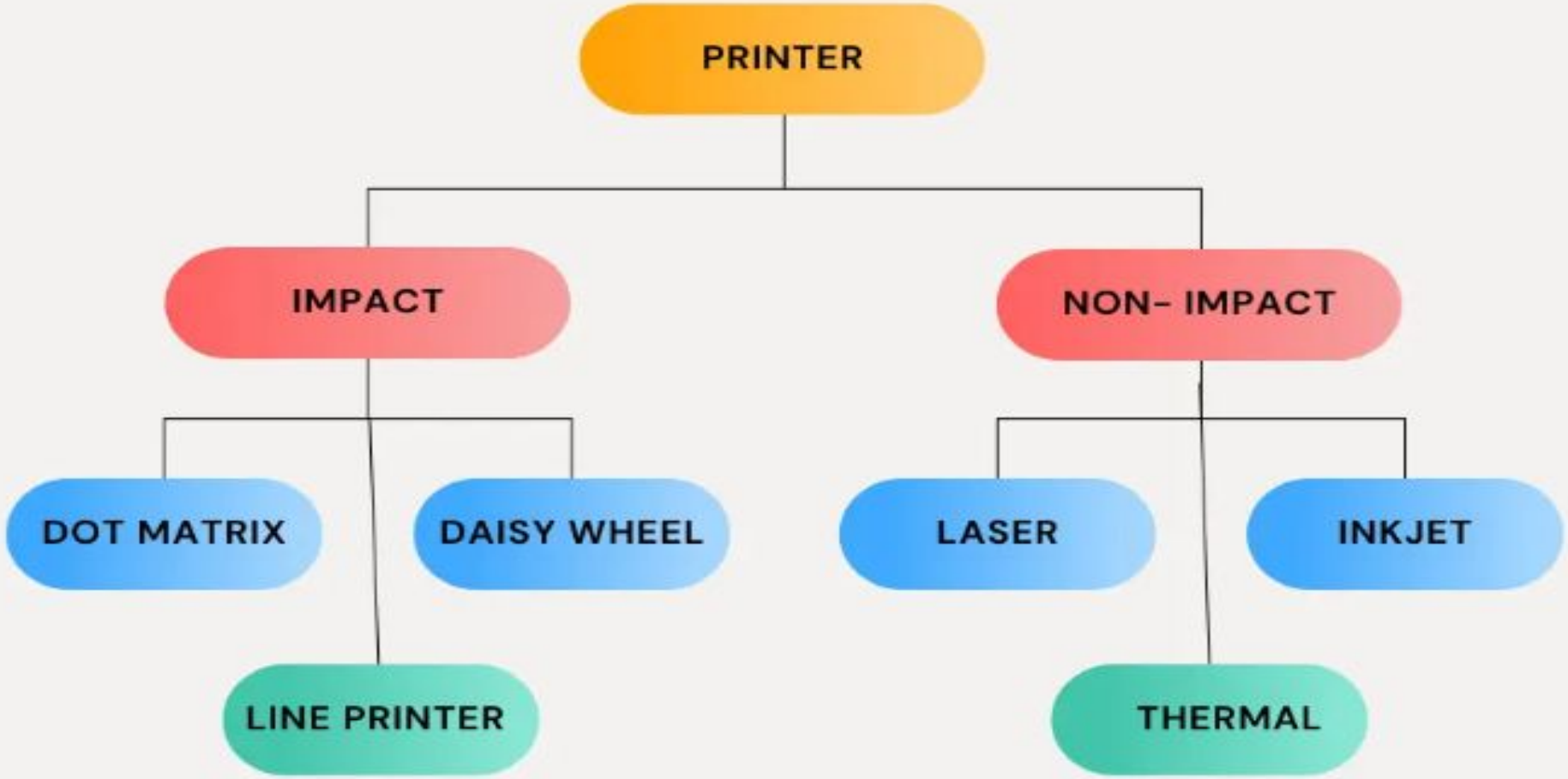
Printers

In the field of computing, a printer is considered a peripheral device that serves the purpose of creating a permanent representation of text or graphics, usually on paper.

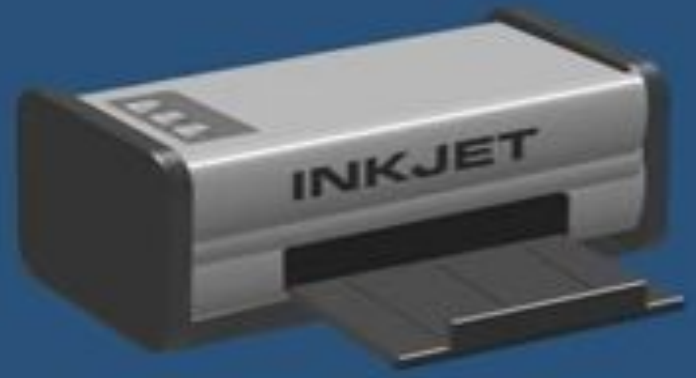
Printers allow the ability to print copies of documents or photos onto paper from a computer

- Printers fall into two general categories: impact and nonimpact.
- Impact printers create an image on paper by using a device to strike an inked ribbon, pressing ink from the ribbon onto the paper. Nonimpact printers use various methods to place ink on the page.

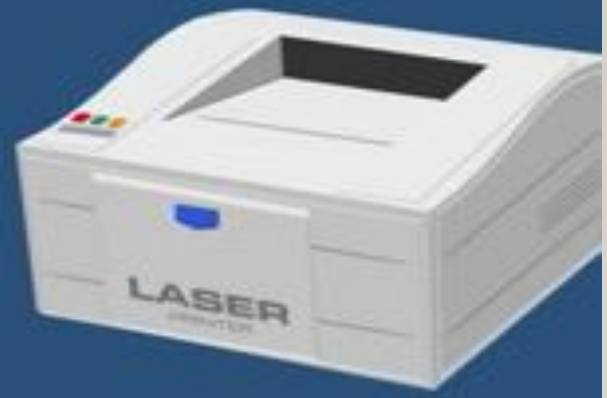
TYPES OF PRINTERS



Printers



MOST COMMON

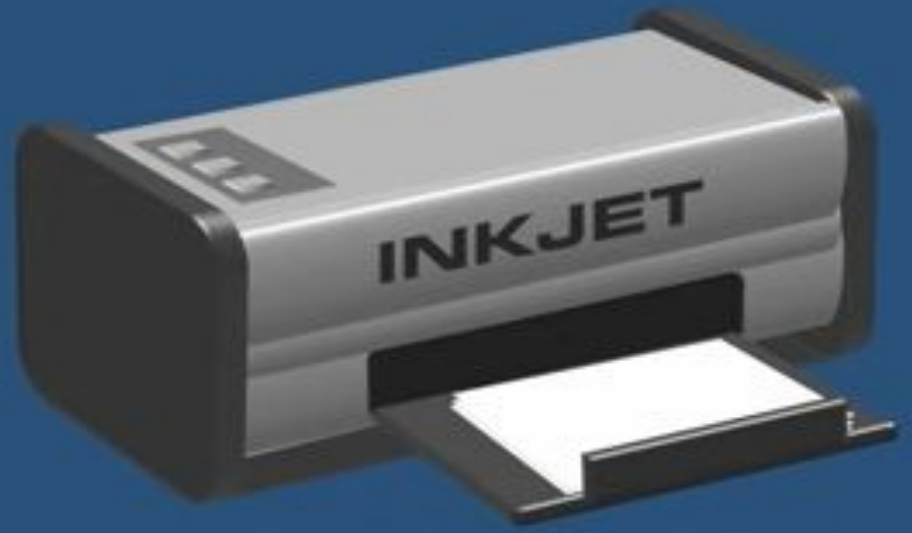


INKJET

LASER

LESS COMMON

Inkjet *PRINTER*

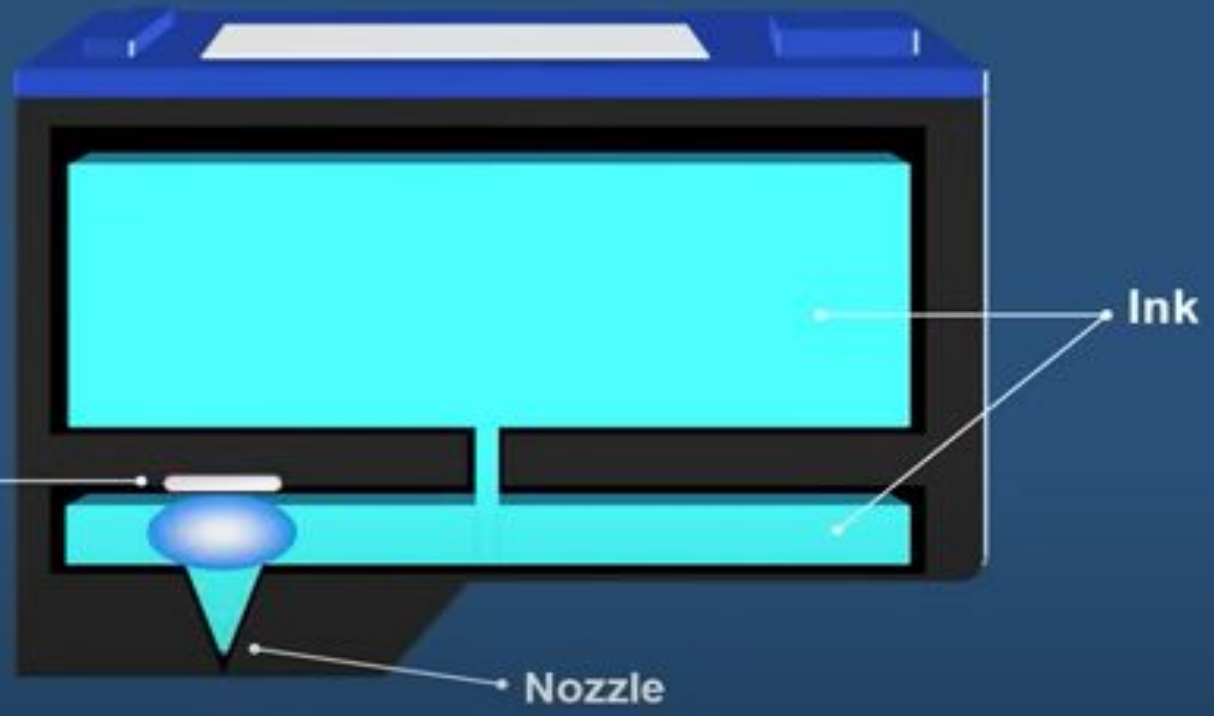


- The most common printers that are used in homes.
- Print head moves back and forth across the paper.
- Print head places ink on paper in very tiny dots.

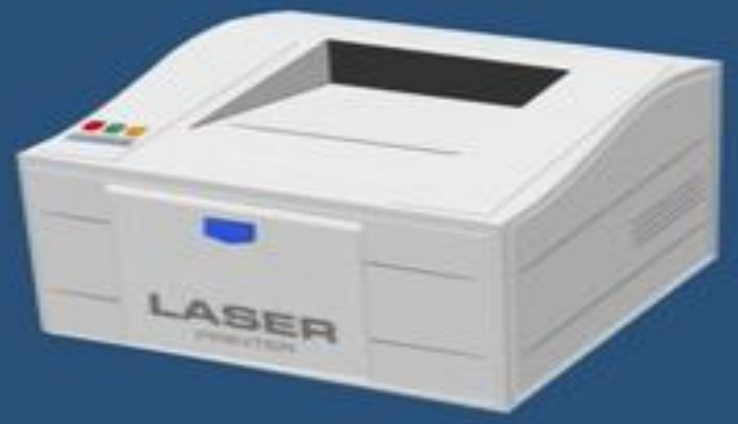
Inkjet *PRINTER*

Thermal Bubble Technology

Used in HP and Canon printers.



Laser *PRINTER*

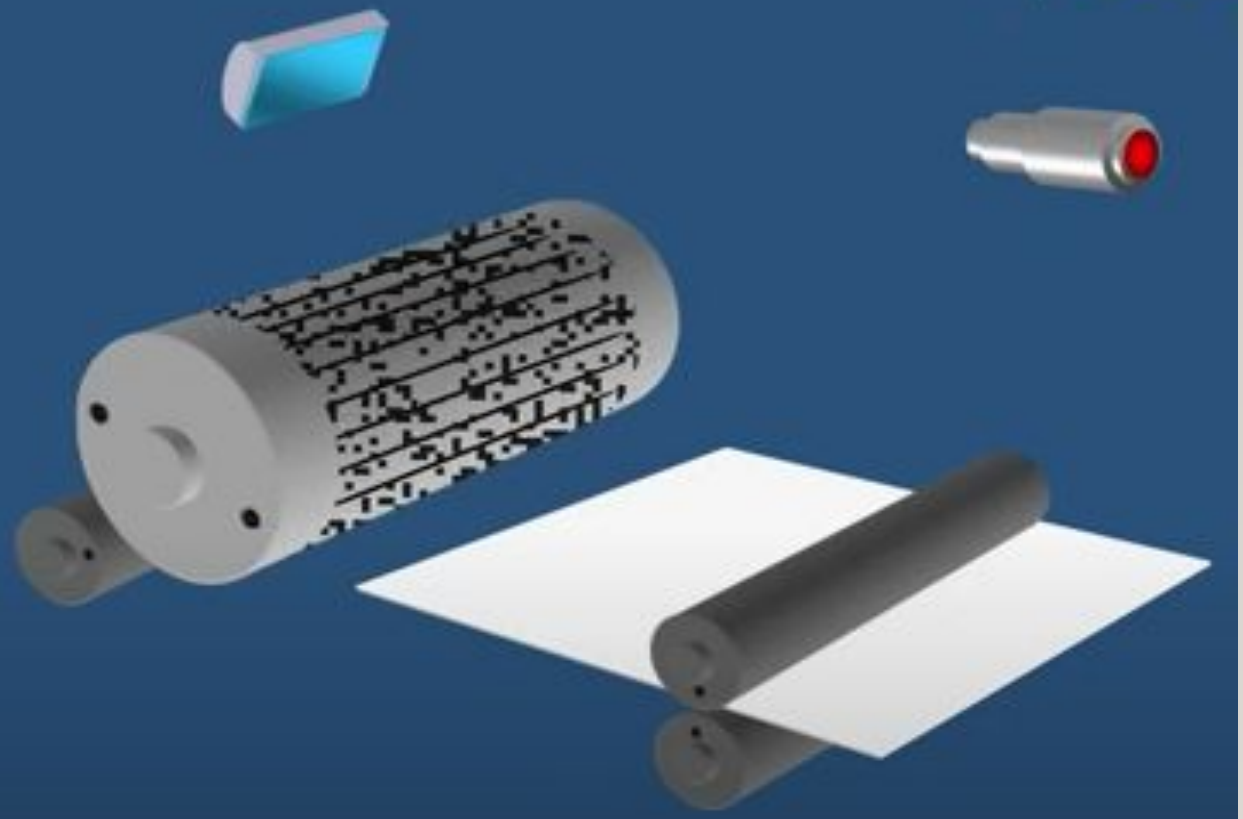


- Laser printers come in different sizes.
- From small personal ones, to larger ones for businesses.
- Provides the highest quality print.
- More expensive than inkjet printers.

Laser *PRINTER*

BASIC OPERATION

- Electric charge is placed on a rotating drum.
- Laser discharges a lower electrical charge on the drum.
- Laser draws the image that is going to be printed on the drum itself.
- Drum is coated with toner.
- Toner only clings to the areas where the laser has drawn.



Thermal *PRINTER*

Finished

Thermal
Paper

Prints by using heat.

Commonly used for printing labels and barcodes.

- A dot matrix printer is an impact printer. It uses a print head that contains a duster of pins. The printer can push the pins out in rapid sequence to form patterns. The pins are used to press an inked ribbon against paper, creating an image.
- An inkjet printer is an example of a nonimpact printer. It creates an image by spraying tiny droplets of ink
- Laser printers produce higher-quality print and are fast and convenient to use, but they are also more expensive than ink jet printers. Laser printers are available in both color and black and white
- Thermal-wax and dye-sublimation printers are used primarily by print shops and publishers to create high-quality color images.
- Photo printers are specialized printers used to print color photographs taken with digital cameras.

Computer Applications in Business

Lecture 2
Looking Inside the computer System

Farjana Rahman Farin
Lecturer, Dept. of CSE, Sonargaon University

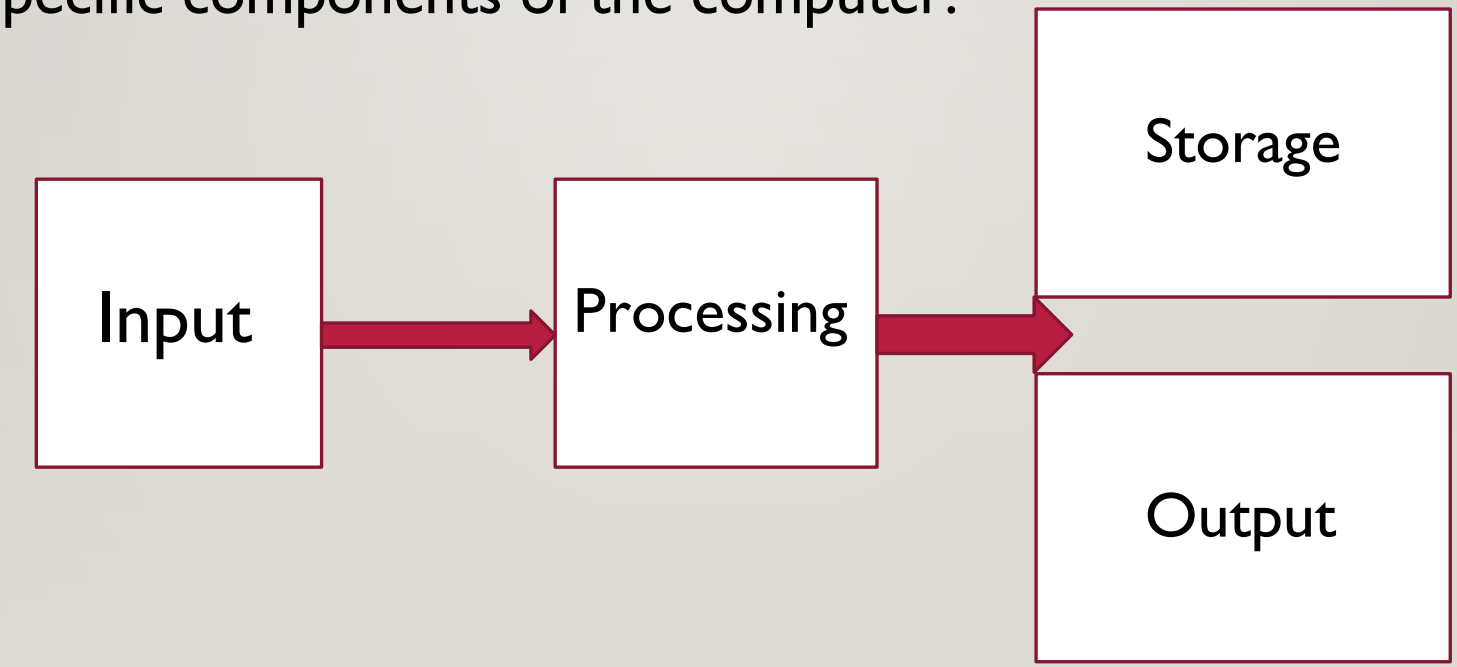
A complete computer system consists of four parts

- ✔ Hardware: The mechanical devices that make up the computer are called hardware. Hardware is any part of the computer you can touch.
- ✔ Software: Software is a set of instructions that makes the computer perform tasks.
- ✔ Data: Data consist of individual facts or pieces of information that by themselves may not make much sense to a person
- ✔ User: People are the computer operators, also known as users

The Information Processing Cycle:

Computer takes data, then processes the data as per being instructed and then can either output the generated information to user or store the information to storage or can do both.

The information processing cycle has four parts, and each part involves one or more specific components of the computer:



Essential Computer Hardware

A computer's hardware devices fall into one of four categories:

1. Processor
2. Memory
3. Input and output
4. Storage

While any type of computer system contains these four types of hardware.

Processor Devices: The procedure that transforms raw data into useful information is called processing. To perform this transformation, the computer uses two components:

- the processor
- memory.

processor is like the brain of the computer; it organizes and carries out instructions that come from either the user or the software

Memory Devices

Memory: memory is one or more sets of chips that store data and/or program instructions, either temporarily or permanently.

- Personal computers use several different types of memory, but the two most important arc called random access memory (RAM) and read-only memory (ROM). These two types of memory work in very different ways and perform distinct functions.

Storage Devices: The purpose of storage is to hold data permanently, even when the computer is turned off.

HW

Computer memory is broadly classified into three categories – **Primary Memory, Secondary Memory,** and **Cache Memory.**

Primary memory	Secondary memory
Primary memory is one that stores data and instructions which are currently being processed by the system or by the CPU.	**Secondary memory** is one that used to store data, programs and information permanently on the computer system.
Primary memory is called the internal memory or main memory.	It is an external memory of the computer .It is also called auxiliary memory or permanent memory.
It is a volatile memory, because when the power is switched off, it loses all its data.	It is a type of non-volatile memory because it retains the data even if the power is switched off.
Primary memory is faster than secondary memory because it is directly accessible to the CPU.	Secondary memory is less faster than primary memory.
The primary memory is further classified into two types namely, **RAM** and **ROM.**	Examples of secondary memories include hard disc, solid state drive, CD, DVD, pen drive, blue ray disc, SD-Card, etc.

Input and Output Devices: Input devices accept data and instructions from the user or from another computer system (such as a computer on the Internet). Output devices return processed data to the user or to another computer system.

- Input Devices:

- **Keyboard :** The keyboard is an essential input device and computer and laptops both use keyboards to give commands to the computer.

- **Mouse:**
  - It is also known as a pointing device.
  - A mouse comprises 3 buttons on the top and one trackball at the bottom which helps in selecting and moving the mouse around, respectively

- **Light Pen:** It is a wand-like looking device which can directly be moved over the device's screen. It is light-sensitive.

- **Microphone:** Using a microphone, sound can be stored in a device in its digital form. It converts sound into an electrical signal.

- **Scanner:** This device can scan images or text and convert it into a digital signal. When we place any piece of a document on a scanner, it converts it into a digital signal and displays it on the computer screen

Input Devices

KEYBOARD

MOUSE

SCANNER

LIGHT PEN

WEB CAMERA

JOYSTICK

TRACK BALL

Output Devices:

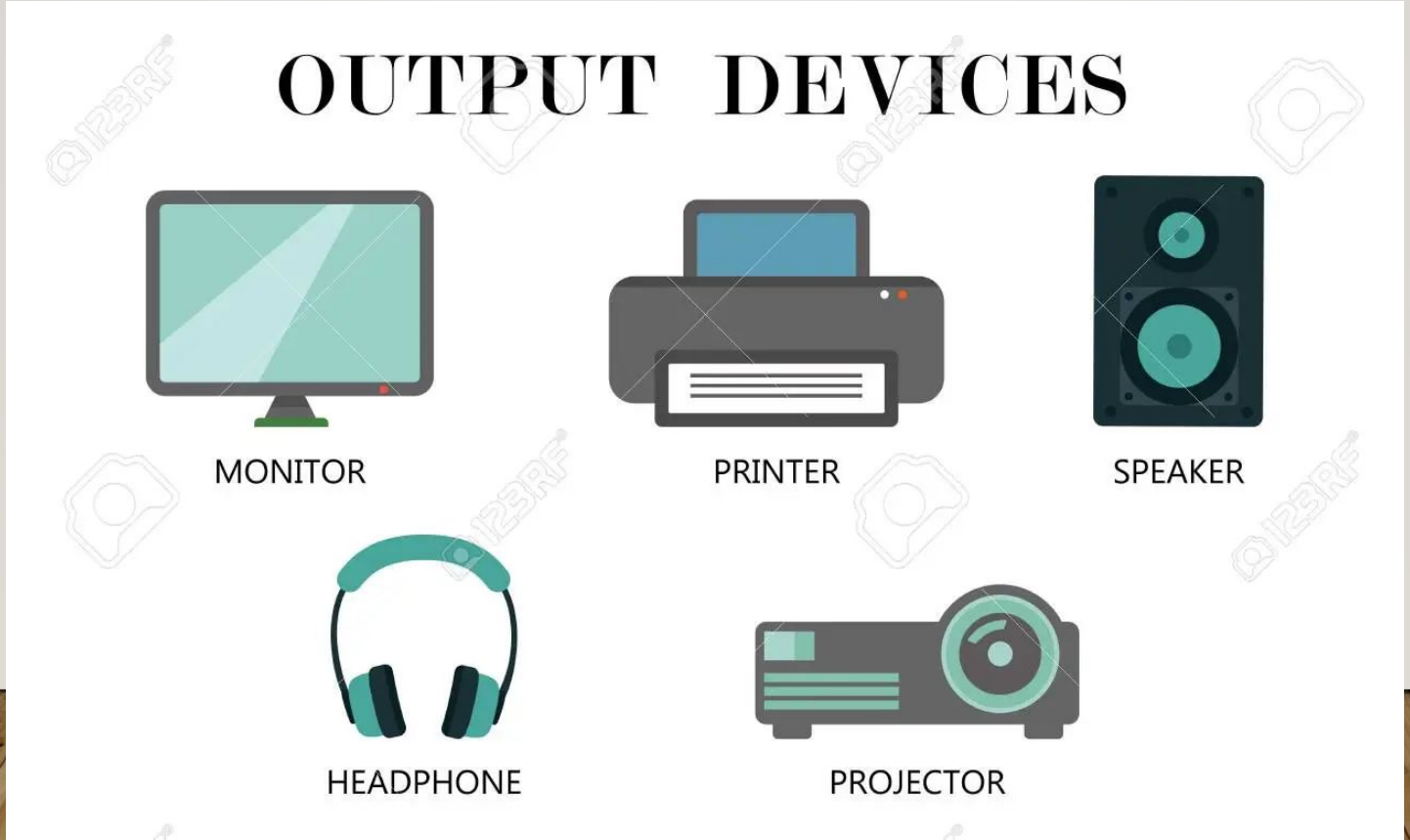
- **Monitor:** The device which displays all the icons, text, images, etc. over a screen is called the Monitor.

- **Printer:** A device which makes a copy of the pictorial or textual content, usually over a paper is called a printer.

- **Speakers:** A device through which we can listen to a sound as an outcome of what we command a computer to do is called a speaker.

- **Projector:** An optical device which presents an image or moving images onto a projection screen is called a projector.

- **Headphones:** They perform the same function as a speaker, the only difference is the frequency of sound. Using speakers, the sound can be heard over a larger area and using headphones, the sound is only audible to the person using them



Software Brings the Machine to Life

The ingredient that enables a computer to perform a specific task is software, which consists of instructions. A set of instructions that drive a computer to perform specific tasks is called a program. These instructions tell the machine's physical components what to do; without the instructions, a computer could not do anything at all.

System Software:

System software is any program that controls the computer's hardware or that can be used to maintain the computer in some way so that it runs more efficiently. There are three basic types of system software:

1. An operating system tells the computer how to use its own components. Examples of operating systems include Windows and Linux. When a program wants the hardware to do something, it communicates through the operating system.

1. A network operating system allows computers to communicate and share data across a network while controlling network operations and overseeing the network's security.
2. A utility is a program that makes the computer system easier to use or performs highly specialized functions.

Application Software:

Application software tells the computer how to accomplish specific tasks, such as word processing or drawing, for the user. Some of the major categories of these applications include.

- Word processing software for creating text-based documents such as newsletters or brochures.
- Spreadsheets for creating numeric-based documents such as budgets or balance sheets.
- Database management software for building and manipulating large sets of data, such as the names, addresses, and phone numbers in a telephone directory.

Computer Use & Users Role

When working with a personal computer; the user can take on several roles, depending on what he or she wants to accomplish:

- Setting up the System.
- Installing Software.
- Running Programs.
- Managing Files
- Maintaining the System

Computer Applications in Business

Lecture 1
Introduction to Computer

Farjana Rahman Farin
Lecturer, Dept. of CSE, Sonargaon University

Contents

- What is Computer?
- Workflow of Computer
- Characteristic features of computers
- Types of Computer's User
- Classification Of Computer

Computer

- The word computer comes from the word “compute”, which means, “to calculate”.

- Thereby, a computer is an electronic device that can perform arithmetic operations at high speed.

- A computer is also called a data processor because it can store, process, and retrieve data whenever desired.

- A computer is an electronic device, that processes data and performs tasks through a combination of hardware and software components, enabling it to execute instructions, store and retrieve information, and solve complex problems. It is made up of two parts- hardware and software.

Simplest workflow Of Computer

- Computer Input is called Data.
- Computer Obtained after Processing data, based on user's instructions is called information.
- Raw facts and figures which can be processed using arithmetic and logical operations to obtain information are called data.
- Processes that can be applied to data are of two types:
  a. Arithmetic Operations.
  b. Logical Operations.

Types of Computer's User

Computers for Individual User:
Personal electronic devices designed for individual use. Example: Laptops, desktop computers, and tablets used by people for tasks like browsing, writing, and entertainment.

Computers for Organizations:
Information processing systems tailored for business or institutional needs. Example: Servers, mainframes, and networked computers used by companies for managing data, communications, and operations.

Computers in Society:
The broader impact and integration of computers in everyday life.
Example: Smartphones, ATMs, and online platforms shaping how people communicate, access information, and participate in the digital age.

Classification Of Computers

All modern Computers and computing devices use microprocessors. Microprocessor speeds and storage capacities are skyrocketing day by day.

Computers are classified on the basis of their use or size:

- Desktop
- Laptop
- Tablet
- Server
- Mainframe
- SuperComputer

Desktop:

Personal Computers (PCs) designed for use by an individual at a fixed location.

- IBM was the first company to introduce and popularize use of desktops.
- A Desktop Unit has

✔ CPU (Central Processing Unit)
✔ Monitor
✔ Keyboard
✔ Mouse

Laptop:

- Compact and portable personal computer introduced in 2000s.
- Enough processing power and storage capacity to be used for all office work, website designing, software development and audio/video editing.
- Laptops run using batteries and connect to networks using Wi-Fi chips.

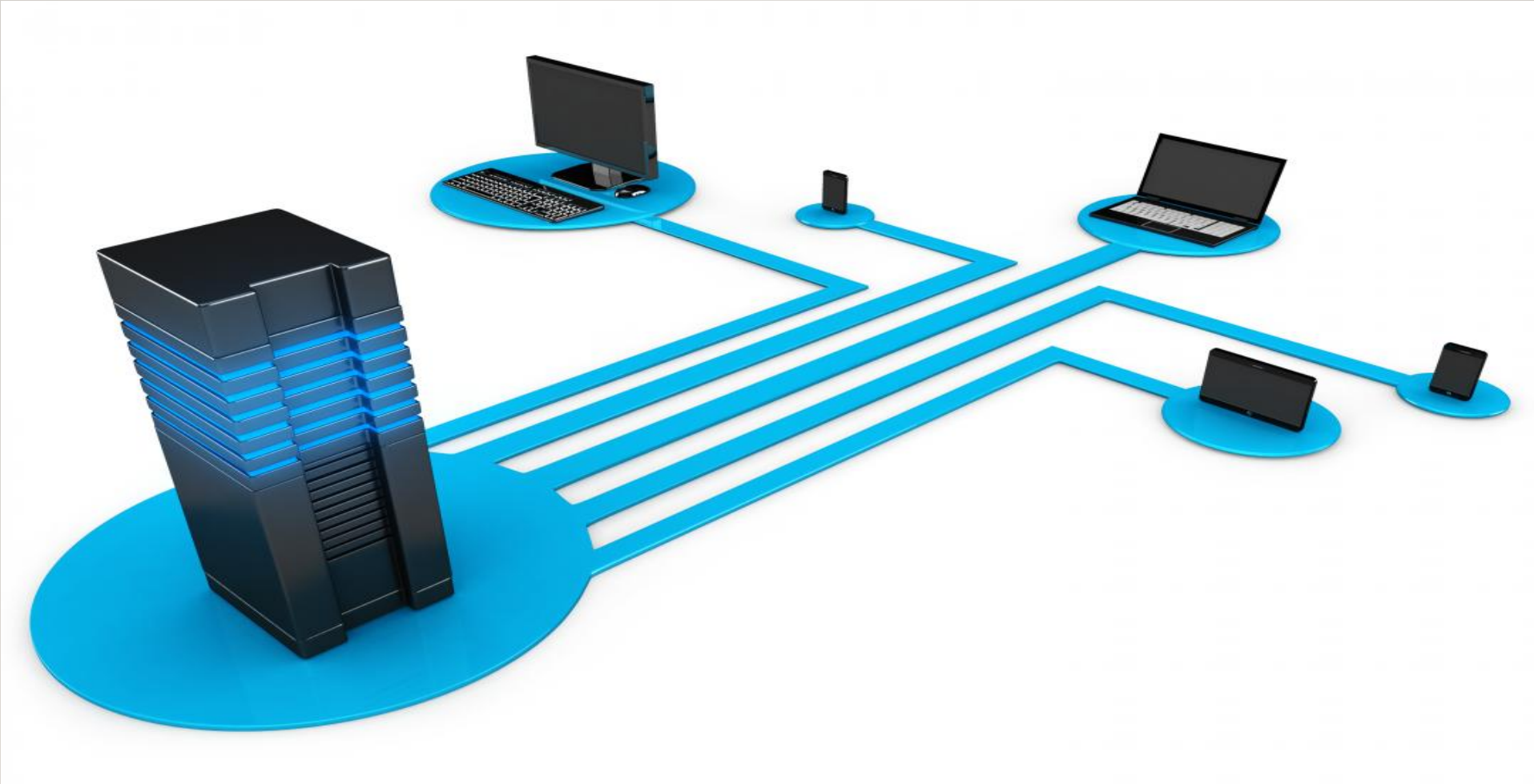
Super Computer:

- Fastest computers on earth
- Used for carrying out complex, fast and time intensive calculations for scientific and engineering applications.
- Speed or performance is measured in teraflops, i.e. 10^12 floating point operations per second.
- Most Common Uses of Supercomputers include:

✔ Molecular Mapping and research
✔ Weather forecasting
✔ Environmental research
✔ Oil and gas exploration

Tablet:

- Processing power of a Desktop.
- Touch sensitive screen of typically 5 to 10 inches.
- Applications that run on tablets are called apps.
- Operating system s provided by Microsoft or Google.



Server

Server:

- **High processing speeds.**
- **Provides one or more services to other systems on the network.**
- **Can handle multiple requests simultaneously.**
- **Most commonly found servers on Networks include.**

❖ **File or storage server**
❖ **Game Server**
❖ **Application Server**
❖ **Database Server**
❖ **Print server**

Main Frame:

- Important features of mainframes are:

- Big in size.
- Hundred times faster than servers.
- Very expensive.
- In built hardware, Software and firmware security features.